1. Overview

Last time, Ian explained how the quantitative zero-free region implies the quantitative prime numbers theorem: if

$$\zeta(\beta + it) = 0 \implies 1 - \beta \gg \frac{1}{\log(2 + |t|)}, \tag{1}$$

then

$$\sum_{n \leq x} \Lambda(n) = x + \mathrm{O}(\exp(-c\sqrt{\log x})) = x(1 + O(\log^{-A} x)). \tag{2}$$

Today we aim to prove the quantitative zero-free region, thereby establishing the quantitative prime number theorem.

As warm-up, we prove a qualitative zero-free region:

$$\zeta(\beta + it) \implies 1 - \beta > 0, \text{ i.e., } \beta = 1. \tag{3}$$

This implies a qualitative prime number theorem

$$\sum_{n \leq x} \Lambda(x) = x + o(x) \tag{4}$$

by the same argument as last time.

2. Proof of the qualitative zero-free region

At the point $s = 1$, the zeta function ζ has a simple pole, so its negative logarithmic derivative $-\zeta'/\zeta$ has a simple pole with residue 1. Thus for $\sigma > 1$,

$$\sum_n \frac{\Lambda(n)}{n^\sigma} = -\frac{\zeta'}{\zeta}(\sigma) = \frac{1}{\sigma - 1} + \mathrm{O}(1). \tag{5}$$

Note that the right hand side tends to ∞ as $\sigma \to 1^+$.

At $s = 1 + it$ with $t \neq 0$, ζ is holomorphic. Let $\ell \geq 0$ denote its order of vanishing. Then $-\zeta'/\zeta$ has a simple pole with residue $-\ell$, and so

$$\sum_n n^{-it} \frac{\Lambda(n)}{n^\sigma} = -\frac{\zeta'}{\zeta}(\sigma + it) = \frac{-\ell}{\sigma - 1} + \mathrm{O}_t(1). \tag{6}$$

Comparing (5) with (6) and using that $\Lambda(n) \geq 0$ and $|n^{it}| \leq 1$, we see that $\ell \leq 1$. We wish to show that $\ell = 0$, so we suppose for the sake of contradiction that $\ell = 1$.

The informal idea is now that (5) and (6) say that when $\Lambda(n) \neq 0$, we have $n^{-it} \approx -1$; but then $n^{-2it} \approx 1$, which contradicts the following consequence of (6) (applied to $2t$ instead of t):

$$\sum_n n^{-2it} \frac{\Lambda(n)}{n^\sigma} \leq \mathrm{O}_t(1). \tag{7}$$

To implement this idea rigorously, we need to make precise the idea that

$$n^{-it} \approx -1 \implies n^{-2it} \approx 1. \tag{8}$$

Let's start simple, with what is essentially the continuity of $z \mapsto z^2$ at $z = -1$:

For each $\varepsilon > 0$ there exists $\delta > 0$ so that $\Re(n^{-it}) < -1 + \delta \implies \Re(n^{-2it}) > 1 - \varepsilon$.

Taking real parts in (5) and (6) gives

$$\sum_{n:\Re(n^{-2it})\leq 1-\varepsilon} \frac{\Lambda(n)}{n^\sigma} \leq \sum_{n:\Re(n^{-it})+1\geq\delta} \frac{\Lambda(n)}{n^\sigma} \leq \frac{1}{\delta}\sum_n (\Re(n^{-it})+1)\frac{\Lambda(n)}{n^\sigma} = \mathrm{O}_t(\frac{1}{\delta}), \quad (9)$$

thus

$$\sum_n (1-\Re(n^{-2it}))\frac{\Lambda(n)}{n^\sigma} = \sum_{n:1-\Re(n^{-2it})\geq\varepsilon} \cdots + \sum_{n:1-\Re(n^{-2it})<\varepsilon} \cdots \quad (10)$$

$$\leq C_t(\frac{1}{\delta} + \frac{\varepsilon}{\sigma-1}), \quad (11)$$

say. Choose $\varepsilon = 1/10C_t$, and take σ close enough to 1 that $\sigma - 1 \leq \delta/10C_t$. Then the above estimate reads

$$\sum_n (1-\Re(n^{-2it}))\frac{\Lambda(n)}{n^\sigma} \leq \frac{2/10}{\sigma-1}. \quad (12)$$

Using (5), it follows that

$$\sum_n \Re(n^{-2it})\frac{\Lambda(n)}{n^\sigma} \geq \frac{8/10}{\sigma-1} + \mathrm{O}(1). \quad (13)$$

But this contradicts (7) upon taking $\sigma \to 1^+$.

This completes the proof of the qualitative zero-free region, hence of the qualitative prime number theorem.

3. Quantitative zero-free region

We need to clean up the above argument and make explicit how everything depends upon t. First, we recall the following estimate from last time, a consequence of factorization theory for holomorphic functions: for $s = \sigma + it$ with $|\sigma| \leq 10$,

$$\zeta(s) = \frac{\prod_{\rho:|s-\rho|<1}(s-\rho)}{s-1}\exp(\mathrm{O}(\mathcal{L})), \quad \mathcal{L} := \log(2+|t|), \quad (14)$$

or equivalently,

$$-\frac{\zeta'}{\zeta}(s) = \frac{1}{s-1} - \sum_{\rho:|s-\rho|<1}\frac{1}{s-\rho} + \mathrm{O}(\mathcal{L}). \quad (15)$$

Note that for $|s-1| \gg 1/\mathcal{L}$ (e.g., for $|t| \gg 1/\mathcal{L}$), we can absorb the first term $\frac{1}{s-1}$ into the error term.

Suppose henceforth that $\sigma > 1$. Since each ρ has real part at most 1, it follows that each fraction $\frac{1}{s-\rho}$ has positive real part when $s = \sigma + it$. Thus for $|t| \gg 1/\mathcal{L}$,

$$-\Re\frac{\zeta'}{\zeta}(\sigma+it) \leq \mathrm{O}(\mathcal{L}). \quad (16)$$

Suppose now that $\zeta(\beta+it) = 0$. We aim to show that $1-\beta \gg \frac{1}{\mathcal{L}}$, with the abbreviation $\mathcal{L}$ as above. Since ζ has a pole at $s=1$, we know that it has no zeros in that region. So we may suppose that $|t| \gg 1$. We improve upon (17) by taking into account the contribution from $\rho = \beta + it$:

$$\sum_n \Re(n^{-it})\frac{\Lambda(n)}{n^\sigma} = -\Re\frac{\zeta'}{\zeta}(\sigma+it) \leq -\frac{1}{\sigma-\beta} + \mathrm{O}(\mathcal{L}). \quad (17)$$

As before, we use (5) to write this as

$$\sum_n (1+\Re(n^{-it}))\frac{\Lambda(n)}{n^\sigma} \leq \frac{1}{\sigma-1} - \frac{1}{\sigma-\beta} + \mathrm{O}(\mathcal{L}). \tag{18}$$

We note also by (5) and (17) (applied to $2t$) that

$$\sum_n (1-\Re(n^{-2it}))\frac{\Lambda(n)}{n^\sigma} \geq \frac{1}{\sigma-1} + \mathrm{O}(\mathcal{L}). \tag{19}$$

We aim now to implement more quantitatively the idea that $n^{-it} \approx -1 \implies n^{-2it} \approx 1$. To that end, let's write $n^{-it} = e^{i\theta}$, so that $\Re(n^{-it}) + 1 = \cos(\theta) + 1$ and $1 - \Re(n^{-2it}) = 1 - \cos(2\theta)$. These quantitaties are related by the formula $\cos(2\theta) = 2\cos^2(\theta) - 1$, from which we deduce that

$$1 - \cos(2\theta) = 2(1-\cos^2(\theta)) = 2(1-\cos(\theta))(1+\cos(\theta)) \leq 4(1+\cos(\theta)), \tag{20}$$

or equivalently,

$$1 - \Re(n^{-2it}) \leq 4(1+\Re(n^{-it})). \tag{21}$$

Inserting this into (18) and (19) gives

$$\frac{1}{\sigma-1} \leq 4\left(\frac{1}{\sigma-1} - \frac{1}{\sigma-\beta}\right) + \mathrm{O}(\mathcal{L}), \tag{22}$$

which may be rearranged as

$$\frac{4}{\sigma-\beta} - \frac{3}{\sigma-1} \leq \mathrm{O}(\mathcal{L}). \tag{23}$$

We choose σ so that $\sigma - 1 = (1-\beta)/\varepsilon$ for some small enough $\varepsilon > 0$; note that if β is close enough to 1, then $\sigma < 10$. Then $\sigma - \beta = (1+1/\varepsilon)(1-\beta)$, so the above estimate reads

$$\frac{1}{1-\beta}\left(\frac{4}{1+1/\varepsilon} - 3\varepsilon\right) \leq \mathrm{O}(\mathcal{L}). \tag{24}$$

Taking $\varepsilon = 1/4$ (say) leads to the required estimate $1 - \beta \gg 1/\mathcal{L}$.

4. How to make the proof seem more mysterious

We can rewrite the key trigonometric inequality as $1 - \cos(2\theta) \leq 4(1+\cos(\theta))$ as $3 + 4\cos(\theta) + \cos(2\theta) \geq 0$, i.e.,

$$3 + 4\Re(n^{-it}) + \Re(n^{-2it}) \geq 0,$$

hence

$$-\frac{\zeta'}{\zeta}(\sigma) + 4\frac{\zeta'}{\zeta}(\sigma+it) + \frac{\zeta'}{\zeta}(\sigma+2it)$$

has nonnegative real part for any $\sigma > 1$, and then integrating gives

$$|\zeta(\sigma)|^3|\zeta(\sigma+it)|^4|\zeta(\sigma+2it)| \geq 1.$$

So we can interpret the above proof in terms of orders of poles, etc.

References